ASRITH POTHURI

Computational Structural Engineer

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Structural engineer with four years of high-rise RCC and steel design experience who builds the automation behind the design workflow — Python tools on the ETABS and Revit APIs for model validation, design automation, and analysis-to-drawing delivery.

EXPERIENCE

Computational Design Engineer — Dar Al Handasah, Bengaluru 08/2025 – Present

- Built and deployed a Python automation framework covering structural modelling, Revit drafting, and QA/QC workflows.
- Developed beam and column tools that map ETABS analysis models to production-ready CAD drawings without manual transcription.
- Built ETABS-to-CAD data linkage for member size verification across thousands of elements.
- Led development of Revit add-ins and ETABS automation tools.

Python · ETABS API · Revit API · AutoCAD automation

Structural Design Engineer II — Walter P Moore, Pune 11/2024 – 07/2025

- Led structural design and analysis of high-rise building systems, applying FEA and advanced analysis methods.
- Maintained internal data APIs synchronising ETABS, Revit, and design documentation systems.
- Coordinated with architects, MEP, and contractors on projects with budgets above $25M.

ETABS · Revit · AutoCAD · Python

Structural Designer — Design Werkz, Pune 07/2022 – 11/2024

- Designed and analysed 30-storey RCC and steel high-rises, including shear walls, outriggers, and transfer structures.
- Performed seismic and wind analysis to IS 875 and IS 1893.
- Ran value engineering studies achieving 15–20% material cost reduction.

ETABS · Revit · AutoCAD · Manual calculations

PROJECTS

ETABS Foundation Auto Tool · Python, ETABS API
Reads column reactions and load combinations, sizes isolated footings against safe bearing capacity, groups them by size, and builds the footing geometry back into ETABS. Resolved 40 columns into 15 footing groups on the demonstration model.

ETABS Property Validator · Python, ETABS API
Audits section properties against their assigned text names across frames, walls, and slabs, and exports a report isolating mismatches. Processes 1,000+ sections in under 30 seconds.

Excel-to-AutoCAD Beam Schedule Generator · Python, AutoCAD automation
Converts Excel reinforcement schedules into dimensioned AutoCAD beam drawings with rebar callouts, layers, and annotation applied automatically.

RHA Concrete Strength Predictor · Python, neural network — M.Tech thesis
Neural network predicting 28-day compressive strength of rice-husk-ash concrete mixes from six mix-proportion inputs, where no code-based prediction method exists. [metric] of [value] against experimental results.

SKILLS

Analysis: ETABS, SAFE, SAP2000, STAAD.Pro

BIM & CAD: Revit (Structure), Revit API, Dynamo, AutoCAD

Programming: Python, VB.NET

Design: High-rise RCC and steel, seismic and wind analysis, shear walls, outriggers, transfer systems, foundation design, value engineering

Codes: IS 456, IS 875, IS 1893

EDUCATION

M.Tech, Computer Aided Structural Engineering — Mahindra University, Hyderabad

B.Tech (Hons), Civil Engineering — Manipal International University, Malaysia

Languages: English, Telugu, Hindi, Malay